

1. Write a brief definition for "Artificial Intelligence"; from the Thinking Humanly view and Acting Humanly view.
2. Draw the diagram of **utility-based** agent
3. Write a brief definition for "Artificial Intelligence"; from the Thinking Humanly view and Acting Humanly view
4. **Define:** An agent - The rational agent - An ideal rational agent
5. **Draw** the diagram and **write** the algorithm of **Model-Based Reflex** agent.
6. Next is a partial tabulation of a simple agent function for the vacuum-cleaner world; write the agent program implements the agent function tabulated.

Percept sequence	Action
[A, Clean]	Left
[A, Dirty]	Suck
[B, Clean]	Right
[B, Dirty]	Suck
[A, Clean], [A, Clean]	Left
[A, Clean], [A, Dirty]	Suck

I. **True or False:**

- A. A Software agent receives network packets as sensory inputs and acts on the environment by sending network packets.
- B. It is possible for a known environment to be partially observable.
- C. If the environment is partially observable, then it could appear to be stochastic.
- D. If the agent has no sensors at all then the environment is partially observable.
- E. In dynamic environments, the current decision could affect all future decisions.

II. **For each of the following sentences; choose the correct answer.**

1. If the next state of the environment not determined by the current state, then we say the environment is

A. fully observable	B. deterministic
C. stochastic	D. static

2. An agent is anything that can be viewed as perceiving its through and acting upon that environment through

A. Sensors – environment – actuators	B.Environment – sensors – actuators
C. Actuators – environment – sensors	D. Environment – actuators – sensors

3. An agent’s behavior is described by an abstract mathematical description and implemented by an

A. agent program	B. agent action
C. agent percept	D. agent function

4. In environments, the current decision could affect all future decisions.

A. continuous	B. dynamic
C. sequential	D. semidynamic

5. The properties of "Sudoku game" task environment:

(Fully Observable - Partially Observable), (Single Agent - Multi-Agents), (Deterministic - Stochastic), (Episodic - Sequential), (Static - Dynamic - Semidynamic), and (Discrete - Continuous)

6. The properties of " Medical diagnosis " task environment:

(Fully Observable - Partially Observable), (Single Agent - Multi-Agents), (Deterministic - Stochastic), (Episodic - Sequential), (Static - Dynamic - Semidynamic), and (Discrete - Continuous).

1. Are the following three statements logically equivalent?

$$\begin{aligned}
 & (p \wedge \neg q) \vee (\neg p \wedge q), \\
 & (\neg p \rightarrow q) \wedge (\neg (p \wedge q)), \\
 & \neg ((p \wedge q) \vee ((\neg p) \wedge (\neg q))).
 \end{aligned}$$